Toxic Language Captures Attention: Evidence from 200 Million News Headline Impressions

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Introduction

Attention has become the fundamental resource that online platforms extract and monetize (Loewenstein and Wojtowicz 2025; Berger, Moe, and Schweidel 2023). Despite remarkable technological developments in media platforms (from newspaper journalism and radio to television, streaming, podcasts and social media), the underlying business model remained remarkably stable: ad-supported platforms provide free content to capture the users' attention, then sell that attention to advertisers. To achieve this, platforms curate their content inventory and typically embed paid advertisements in ways that are difficult to ignore (Wu 2017). This model evolved as user-generated content expanded content inventories and required automated content delivery systems that maximize attention and advertisement revenue (Aridor et al. 2024). Where traditional media outlets employed human editors to curate content, modern platforms deploy data-driven recommender systems that determine which content is displayed to whom. However, these algorithms amplify whatever content attracts the most attention, regardless of its social value: since users are often drawn to content that is not only engaging but also toxic (e.g., hate speech, harassment, and related forms of content, see Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024), platforms may inadvertently promote content that degrades public discourse and the users' well-being (Guess et al. 2023; Bursztyn et al. 2023).

Analyzing 53,755 news headlines in a unique setting of 12,473 A/B-tests that generated over 205.8M impressions, I find that toxic content systematically outperforms civil alternatives

in capturing clicks. Specifically, toxic content increases click-through rates (CTR) by 9.859% even when advertising identical articles.

These findings reflect deeper psychological mechanisms that characterize the platforms' "battle for clicks and eyeballs" (Senior 2016). In competitive information environments where each piece of content only has (milli-) seconds to attract a user's attention, content can be curated to stand out (e.g., because of visual contrasts, social and personal significance, or emotional significance) and capture attention before any deliberate evaluation occurs. Hence, platforms can leverage so-called bottom-up mechanisms that operate outside of a user's volitional control (Loewenstein and Wojtowicz 2025) and arise in response to properties of their stimuli to capture and, ultimately, steer their users' attention.

Evidence from multiple disciplines suggests that negative content (Robertson et al. 2023; Banerjee and Urminsky 2024) may be particularly effective at capturing attention through these automatic processes. Baumeister et al. (2001) suggest that an attraction to such content may be deeply rooted due to evolutionary origins: organisms that were better attuned to bad things would have been more likely to survive threats (see also Öhman and Mineka 2001). Accordingly, toxic language may trigger threat-detection mechanisms even when the "threats" are manufactured for a platform's monetization rather than a user's survival (Rathje and Van Bavel 2025).[1]

Prior literature on toxicity not only found negative effects on welfare but also positive effects on engagement (Beknazar-Yuzbashev et al. 2022; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski 2024). However, conceptualizing engagement faces a methodological challenge: to disentangle toxicity's direct effect on individual content consumption from confounding social dynamics. Because such evidence comes from social media platforms, the reported engagement decisions were public and may have served social signaling functions (Huang and Lin 2025) rather than information seeking. To address this concern, I analyze private click

[1] In fact, recent studies suggest that there is a disconnect between the users' and the media platforms' welfare: while Beknazar-Yuzbashev et al. (2022) show that toxic content may be profitable, other experimental evidence also suggests that active social media users would prefer platforms such as TikTok and Instagram not to exist (Bursztyn et al. 2023). Rathje and Van Bavel (2025) term this disconnect the *"paradox of virality"* and compare the engagement with toxic content to rubbernecking at car accidents. People feel compelled to look even when they consciously wish to look away.

decisions in a field experimental setting where users click on headlines solely to access article content, with no immediate social visibility or signaling opportunities.

The remainder of this paper proceeds as follows. The next section describes the Upworthy Research Archive dataset, explains the toxicity measurement approach using the Perspective API, and discusses click-through rates as the primary outcome variable. Next, I outline the empirical strategy that exploits within-experiment variation to identify the association between toxicity and engagement. I then present main findings and robustness checks, followed by a discussion of the implications for content optimization and platform design. The paper concludes with directions for future research.

Data

The Upworthy Research Archive

This paper analyzes variations of news headlines written by writers and editors of Upworthy.com, a U.S. media publisher that was recognized as the fastest-growing media company in the world in 2013. Upworthy pioneered the systematic use of randomized A/B-tests (Matias et al. 2021), hereafter referred to as *"experiments"*. For each article, Upworthy tested different variations of headlines to advertise the respective article on its home feed. The corresponding experimental process was straightforward: editors or writers created several alternative headlines per news article, website visitors were randomly assigned to one version upon visiting Upworthy.com, and the platform recorded whether the visitor clicked, producing a binary outcome for each headline. The headline with the most clicks per impression (referred to as click-through rate, CTR) was then selected to advertise the article permanently and for all users. Because these experiments were not only conducted systematically, but also deployed at scale (i.e., at a time when Upworthy's content was shared more frequently on Facebook than that of all major U.S. outlets combined) this case is well-suited to study attention allocation and data-driven content optimization.

The *Upworthy Research Archive* (Matias et al. 2021) retrieved the corresponding data and contains all experiments conducted between January 24, 2013, and April 30, 2015. For this

study, I use the cleaned version of the dataset provided by Reiss and Roggenkamp (2025), who removed duplicate headlines and added additional attributes (e.g., semantic negativity scores predicted by gpt-4o-mini). The resulting subset includes 12,473 experiments with 53,755 unique headline variations, that generated approximately 205.8M impressions and 2.8M clicks.

The dataset has a two-level structure: the article level corresponds to an experiment (A/B-test), and the headline level, where multiple alternatives advertise the same article. Table 1 displays six alternative headlines that advertised the same underlying article to illustrate an experiment's nested data structure. The headlines exhibit substantial variation in both toxicity scores (ranging from .044 to .859) and click-through rates (ranging from .343% to .866%).

Table 1: Headlines from an examplary A/B-test describing the same article

Headline	Toxicity	CTR (%)
Here's How You End Up In Prison If You're British And Also A Complete Idiot	.859	.866
He's The Dumb Husband. She's A Doting Wife. And Then Something Important Landed At Their Door.	.627	.674
What They Learned Will Change What They Think About The Planet. (But They'll Still Be British.)	.112	.343
In Just 1 Minute, These Two People Learned The One Comment They're Not Supposed To Make	.109	.596
A Couple That Will Never Make The Same, Simple Mistake Again	.106	.493
An Incredibly Important Piece Of Information Came Right To Their Front Door	.044	.350

Note: Toxicity scores range from 0 to 1 and represent the probability that human raters would classify the content as toxic, as measured by the Perspective API. CTR = Click-Through Rate (clicks/impressions). Headlines are ordered by toxicity score (highest to lowest). All headlines advertised the same underlying article.

In contrast to online platform and observational studies (see Roggenkamp, Boegershausen, and Hildebrand 2025), the data were not affected by algorithmic interference issues (see, e.g., Xu, Zhang, and Zhou 2020; Johnson 2023; Braun and Schwartz 2025). In addition, Upworthy conducted only one experiment at a time across their entire website to ensure that the tests were independent.

Matias et al. (2021) collaborated and interviewed with former engineers, editors, and data scientists employed at Upworthy to document and validate the dataset structure. All seven interviewees confirmed details about the experimental process and data collection procedures. An additional quantitative validation checked balance across experimental arms and confirmed that clicks were always lower than or equal to participant counts.[2]

Toxicity Prediction

To measure toxicity in the Upworthy headlines, I employed the Perspective API (Lees et al. 2022; Votta 2025), a machine learning system developed by Jigsaw (a unit within Alphabet Inc.) for identifying toxic content in online conversations. The API was released in 2017 and has been widely adopted by major publishers including The New York Times, Le Monde, and The Financial Times for content moderation purposes.

The underlying models are trained on millions of comments from a variety of sources, including Wikipedia and The New York Times, across several languages. For each comment, 3-10 human annotators evaluated whether the comment contains toxicity following standardized instructions. As a result, the API generates toxicity scores on a continuous scale from 0 to 1, that represent the probability that a given text would be perceived as toxic by human raters. For this reason, the scores have a direct interpretation based on the training methodology: a score of 0.7 suggests that 7 out of 10 raters would classify content as toxic. The API's definition of toxicity, therefore, is rooted in the training instructions provided to human annotators. Specifically, they evaluated the toxicity in each comment, with "toxic" content defined as "a

[2]The validation process identified treatment imbalance issues for tests conducted between June 25, 2013, and January 10, 2014, leading to the exclusion or special treatment of experiments from this period in subsequent analyses.

rude, disrespectful, or unreasonable comment that is somewhat likely to make you leave a discussion or give up on sharing your perspective" (Lees et al. 2022).

For the analysis, I processed each headline in the Upworthy dataset through the Perspective API to obtain toxicity scores. The API employs character-level transformer models that are well-suited for analyzing news headlines, as they can handle the varied linguistic styles and creative language common in attention-grabbing content. With an Area Under the Curve (AUC) of .974 for toxicity detection (Lees et al. 2022), the API provides both strong empirical performance in addition to its ecological validity (given that it is deployed by major publishers for real-world content moderation).

Despite predicting continuous likelihood scores, I also use the predictions to flag (and classify) headlines that exceed a threshold of .700. This threshold represents a critical benchmark in content moderation practices that typically triggers moderation actions such as reduced visibility or removal (see also Horta Ribeiro, Cheng, and West 2023). I thereby follow the Perspective API's recommendation[3] to use thresholds between 0.7 and 0.9 to mirror typical industry moderation use cases.

The predicted toxicity scores range from .000 to .929. Yet, only 295 (.623%) reached the recommended minimum threshold of .700. Hence, the corresponding distribution is right-skewed as displayed in Panel A of Figure 1. The scarcity of toxic headlines likely reflects the editorial and reputational constraints of a professional media outlet. As a result, the claims derived from this dataset may best be extrapolated to other, potentially user-generated contexts, that a characterized by predominantly civil discourses.

Even though the Perspective API represents state-of-the-art technology for automated toxicity detection, several limitations must be acknowledged. The models may be sensitive to the presence of profanity or certain keywords, potentially flagging content that uses strong language but is not necessarily harmful in context. Additionally, the API's performance varies across different types of content and may exhibit biases related to the demographic characteristics of the original training annotators or the specific domains represented in the training data. However, for research purposes examining patterns of toxicity in news headlines, the API pro-

[3] https://developers.perspectiveapi.com/s/about-the-api-score, accessed 2025-08-12

vides a consistent measurement approach that enables large-scale analysis while maintaining interpretability through its probabilistic framework.

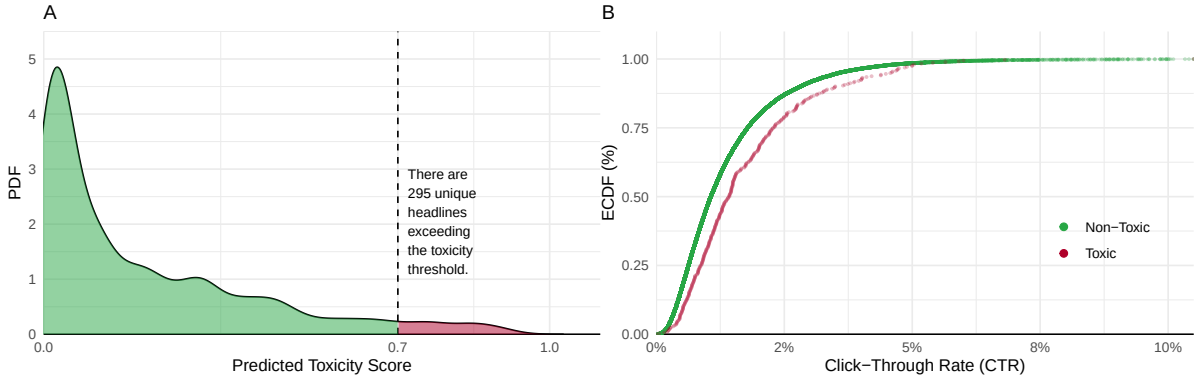
Click-Through Rate

Throughout this paper, the click-through rate (CTR), calculated as the ratio of clicks to impressions for each headline variation, is the primary dependent variable. Because users click on a headline to reveal more information related to that headline, the CTR offers several advantages for measuring attention allocation. First, a click represents behavioral outcome that requires minimal but non-trivial effort. As a consequence, the CTR captures users' intention to acquire information rather than passive exposure. Second, clicks are consequential as they reflect trade-offs with opportunity costs: choosing to learn more about one headline means forgoing alternatives (Reutskaja et al. 2011; Schütz and Stewart 2025). Third, the binary nature of clicks aligns well with the research question about whether toxic language influences attention capture. I am specifically interested in the initial (bottom-up) decision to acquire more information (i.e., the moment where headline characteristics translate into user behavior) rather than subsequent reading time or content consumption (Berger, Moe, and Schweidel 2023). Finally, clicks also represent the actual optimization metric that Upworthy used to guide editorial decisions. This is aligned with the information recommender systems process and implies that the findings speak directly to the incentives and trade-offs content creators and platform algorithms face more generally.

Overall, the recorded CTRs range from .000% to 14.891% and the distribution of click-through rates is right-skewed: most headlines attract relatively little attention, but a small fraction achieve high CTRs.[4] This long-tail distribution (see Panel B in Figure 1) mirrors patterns of virality in digital media, where a few pieces of content disproportionally attract attention. It also means that effect sizes expressed in percentage points should be interpreted relative to a low baseline median CTR of 1.067%.

[4]Only .072% of all headlines generated CTRs above 10%.

Figure 1: Distribution of predicted toxicity scores and click-through rate performance



Distribution of Predicted Toxicity Scores Across Headlines, A. The density plot shows that most headlines receive low toxicity scores, with only 295 headlines (.623% of the sample) exceeding the .700 threshold for toxic content. **Empirical Cumulative Distribution of Click-Through Rates by Toxicity Classification, B.** Headlines exceeding the toxicity threshold (red) systematically achieve higher click-through rates than non-toxic alternatives (green), with the toxic distribution shifted rightward across the entire range of performance outcomes.

Empirical Strategy

Panel B of Figure 1 provides initial, model-free evidence of a relationship between toxicity and engagement. The empirical cumulative distribution functions illustrate that headlines flagged as toxic consistently achieve higher click-through rates than civil alternatives, with the toxic distribution shifted to the right. This raw pattern suggests that toxic content captures more attention across the entire distribution of the CTR. However, this model-free comparison may be misleading if toxic headlines systematically advertise different types of articles than non-toxic headlines. Articles that naturally attract toxic language (such as controversial political topics) might also inherently generate higher CTRs regardless of headline phrasing. To better capture the effect of toxic language from these article-level characteristics, I exploit variation in headline toxicity within experiments to estimate whether CTR increases in toxicity when comparing headlines that advertise identical content.

To this end, I employ a fixed effects specification that controls for all article-level characteristics:

$$\text{CTR}_{ij} = \alpha_i + \beta \cdot \text{Toxicity}_{ij} + \gamma' \mathbf{X}_{ij} + \varepsilon_{ij}$$

where CTR_{ij} is the click-through rate for headline j in experiment i, Toxicity_{ij} is the Perspective API's toxicity score (0-1), and α_i are experiment fixed effects. The vector $\mathbf{X}_{ij}$ includes headline-level controls: all models include headline length (word count) and text complexity (Gunning Fog Index) as covariates since Robertson et al. (2023) found these to predict CTR.[5]

The coefficient of interest, β captures the association between toxicity and CTR using only variation across headlines for the same article. For example, this approach compares the following two headlines advertising the same article reported in Table 1: *"An Incredibly Important Piece Of Information Came Right To Their Front Door"* with *"Here's How You End Up In Prison If You're British And Also A Complete Idiot"* (toxicity scores: .044 vs. .859) while holding constant the underlying article content and all article-invariant factors.

The remaining identification challenge is that editors have not systematically varried toxic language within their A/B-tests. Instead, variation in toxicity occurs naturally and may have been paired with other engagement tactics. If editors systematically bundled toxic language with specific engagement tactics within the same article, β captures the combined effect rather than toxicity's isolated impact. Banerjee and Urminsky (2024) note that in their review of the prior academic literature on linguistic drivers of engagement, academic experiments designed to test a single construct typically also included both intended and incidental differences because of inherent linguistic limitations that often required more changes than the intended change to preserve message realism (see also Simonsohn, Montealegre, and Evangelidis 2025) In addition, the scale of the dataset (which spans 12,473 experiments created by multiple editors over two years) exhibits heterogeneity in editorial approaches. This heterogeneity means toxic language is likely paired with different engagement tactics across experiments rather than systematically bundled with the same strategies, allowing β to better capture toxicity's effect on CTR.

[5]Platform age, another covariate they controled for in their mixed effect models, varies only between experiments, so article fixed effects absorb this variation.

Results

The baseline specification (see Model 1 in Table 2) reveals a positive association between toxicity and engagement: a one standard deviation increase in toxicity corresponds to a .045 percentage point increase in CTR (SE = .003, z = 14.907, 99% CI = [.037, .053]).

Model 2 tests whether toxicity's effect persists when controlling for negativity (as measured in Reiss and Roggenkamp 2025). The minimal change in toxicity's coefficient (from .045 to .042) when adding negativity suggests that toxicity adds predictive power beyond negativity (SE = .003, z = 13.669, 99% CI = [.034, .050]). Consistent with this interpretation, the two measures are only moderately correlated ($\rho_{Pearson}$ = .242), which suggests partial overlap but also distinct variance. Users therefore appear drawn both to negative content (bad news, unfortunate events) and to hostile or disrespectful language, with these effects being additive rather than redundant. This pattern indicates that toxicity captures something beyond mere negativity, such as hostile, disrespectful, or threatening elements that may trigger automatic threat-detection responses. Importantly, both toxicity and negativity show, by and large, comparable positive associations with CTR, which underscores that each has a meaningfull effect on CTR.

Model 3 addresses a measurement concern noted in the Perspective API documentation: the algorithm may conflate profanity with toxicity, potentially misspecifying content that relies on strong language as toxic. As a consequence, there is a high correlation between toxicity and profanity ($\rho_{Pearson}$ = .865), which requires a careful interpretation. When both are included, toxicity's coefficient increases to .059 percentage points (SE = .006, z = 10.153, 99% CI = [.044, .074]) while profanity shows a negative association (b = -.019, SE = .005, z = -3.519, 99% CI = [-.032, -.005]). Rather than indicating that profanity reduces engagement, this pattern likely reflects the multicollinearity between these measures as the model cannot reliably separate their individual contributions. Hence, the most conservative interpretation focuses on Model 2, which suggests that toxic language, irrespective of the level negativity, is associated with higher engagement rates.

To contextualize these effect sizes: if the most toxic headlines in the dataset (scoring above

0.7) were compared to civil alternatives for the same articles, the model predicts approximately 9.859$ higher click-through rates. However, this extrapolation assumes linearity across the full toxicity range, which may not hold given the limited variation in highly toxic content within this professionally edited dataset.

Table 2: Fixed Effects Specifications

	Model 1		Model 2		Model 3	
	b	99% CI	b	99% CI	b	99% CI
Toxicity	0.045	0.037, 0.053	0.042	0.034, 0.050	0.059	0.044, 0.074
Negativity			0.067	0.034, 0.100	0.061	0.028, 0.094
Profanity					-0.019	-0.032, -0.005

N = 53,755 headline variations across 12,473 experiments. Dependent variable: click-through rate (mean = 0.014). b = coefficient estimate (in percentage points) with 99% confidence intervals. All coefficients significant at $p < 0.001$. All models include article fixed effects and controls for headline length and text complexity. All coefficients are z-standardized. Models weighted by impressions.

Discussion

This research contributes to the emerging literature on toxic content and user behavior (Beknazar-Yuzbashev et al. 2022) by providing evidence that toxic language systematically captures more attention than civil alternatives. By using a large-scale experimental dataset that isolates toxicity effects from social signaling concerns, I conceptually replicate and extend prior findings. As I find a toxicity bias while focusing on attention (instead of engagement) the findings suggest a mechanism grounded in evolutionary psychology and attention research.

- Engagement-based optimization creates systematic selection pressure favoring toxic content, even in professional editorial contexts where toxicity is rare and constrained.
- My estimates reflect associations between toxicity and engagement within the set of headlines that editors chose to test. If editors systematically paired toxic language with other engagement tactics, our estimates may confound toxicity effects with these other factors.
- Our results may be interpreted as the 'total effect' of using language that gets flagged as toxic in real-world content optimization, which includes both direct toxicity effects and correlated engagement tactics.
- Toxicity scores of 1 do not exist in my sample. hence, reporting effects where raters would unanimously agree on a headline being toxic (i.e., `TOXICITY == 1`) is a little misleading. Consider reporting effects crossing the threshold instead.
- The paper applies this logic to "platforms" generally, but your study focuses on one specific platform (Upworthy) and one specific type of content (news headlines). The broader generalization might be stronger than your evidence supports.
- Further research should use DICE (Roggenkamp, Boegershausen, and Hildebrand 2025) because the effects are likely not linear (Vettehen, Nuijten, and Peeters 2008). Also DICE would allow to measure the toxicity of feed compositions. Not only of headlines.

- (Berger and Milkman 2012) used older data from NYT and analyzed whether they make it on the most emailed list (another social construct in a high-reputation context). They find that positive word drive virality. They also show that more affect-laden content, regardless of valence, drives engagement. While I cannot replicate the former finding, I replicate the latter through toxicity.

- These findings relate to platform analytics and brand safety as they suggest that (social) media platforms may promote harmful content (i.e., brand-unsafe environments for their ads), triggering question of equilibria in presence./absence of such concerns.

- Despite some links to social media, this research has implications for other channels such as eMail marketing or push notifications where recipients (not unlike news headlines) the subject lines (and some lede).

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Appendix

Robustness Checks

Mixed Effects Models Specification

Clustered Standard Errors Specification

Legacy